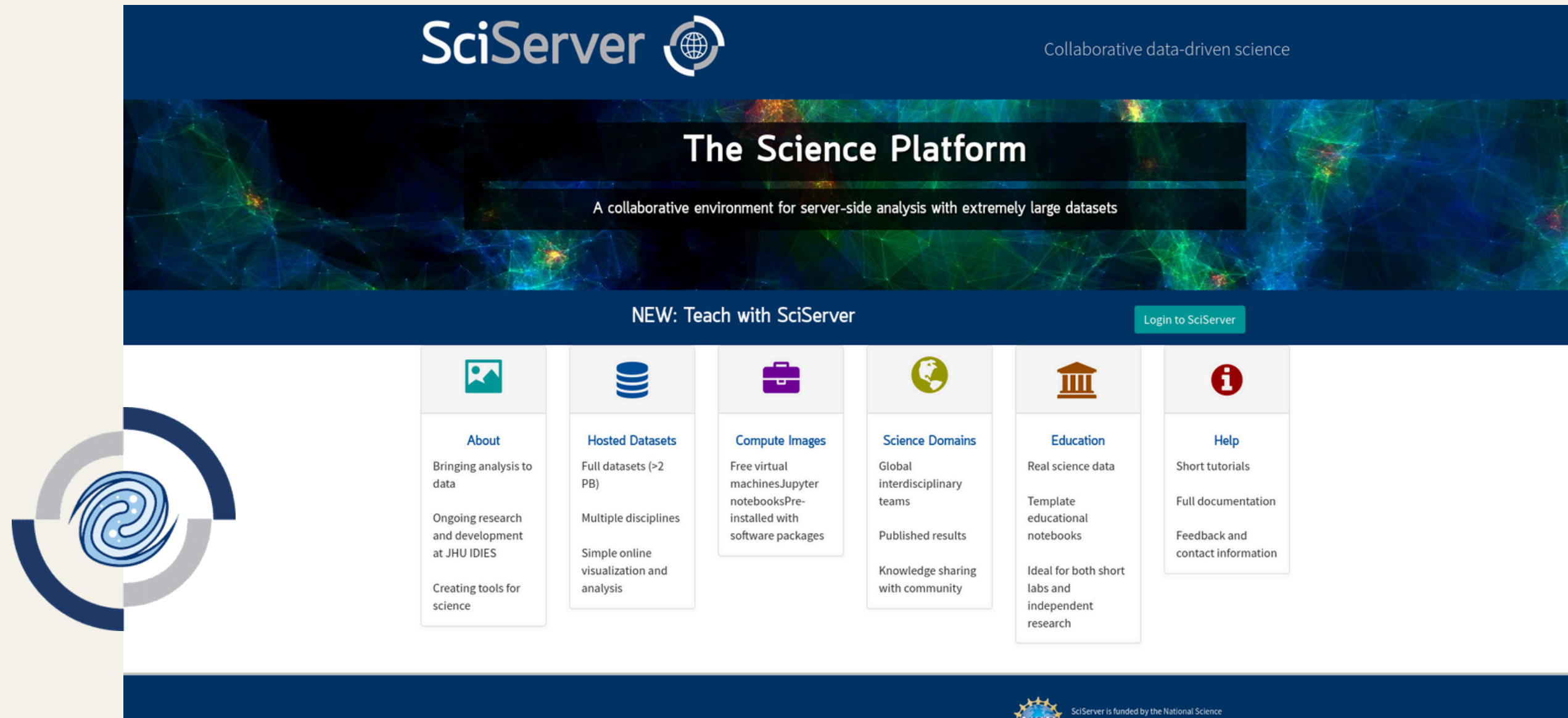




Collaborative data-driven science



SciServer is a cloud-based data analysis platform designed by Johns Hopkins University in collaboration with NASA

<https://www.sciserver.org/>

SciServer Dashboard

Data, Collaboration, Compute

Your Activities



Files

You have 1 Shared User Volume.
You have 2 Owned User Volumes.



Groups

You have 0 Group Invitations.
You have 1 Owned Group.



Compute Jobs

You have 0 Jobs Running.
You have 0 Jobs Completed in 24 hours.



Science Domains

You have joined 0 domains.
There are 3 domains available.

SciServer Apps



CasJobs

Search online big relational databases collection, store the results online, and share them.



Compute

Analyze data with interactive Jupyter notebooks in Python, R and MATLAB.



Compute Jobs

Synchronously run Jupyter notebooks in Python, R and MATLAB or commands.



SkyServer

Access the Sloan Digital Sky Survey data, tutorials and educational materials.

Containers

Created At	Name	Domain	Image	Status	
2025-05-20 10:07:06.0	SWIFT2	Interactive	SciServer Essentials 4.0	running	  
2025-05-20 09:32:12.0	SWIFT	Interactive	SciServer Essentials 4.0	running	  
2025-04-04 19:57:38.0	XMM_TWO	Grendel	HEASARCV6.35	running	  

Create container

Important Information about Compute Container File Storage

File System

Most of the folders in a Container's file system should not be used to store your files. Your initial container view is of `/home/idies/workspace`, which may contain volumes under the `Storage` and `Temporary` folders. Any user volumes you choose to add to the container at creation will be present within these folders. Do not store your files in `workspace`, or in any other folder except as described here. If a Compute node fails, your incorrectly stored files will be *lost permanently*.

Storage

Use `Storage` volumes for long term storage of your scripts and small data files. The volumes in the `Storage` folder are backed up. These volumes are mounted according to the username of the user who created them under the path `/home/idies/workspace/Storage/username/user volume name/`. Files saved to this folder persist between your containers, even in the event that a container fails. Other files and folders cannot be placed in any intermediate paths, i.e., under the `Storage/username` or `Storage/` folders. Your `Storage` volumes are subject to size limitations described in the [SciServer Compute Data Storage Policy](#).

persistent

By default, all users start with a `Storage` volume named `persistent`. The files in this volume correspond to the same `persistent` folder used in previous versions of Compute.

Temporary

Use `Temporary` volumes for temporary large file storage. The `Temporary` volumes persist between containers and are not affected by Compute node failure, but is not backed up. These volumes are mounted according to the username of the user who created them under the path `/home/idies/workspace/Temporary/username/user volume name/`. Other files and folders cannot be placed in any intermediate paths, i.e., under the `Temporary/username` or `Temporary/` folders. Your `Temporary` volumes are subject to time limit and size limitations described in the [SciServer Compute Data Storage Policy](#).

scratch

By default, all users start with a `Temporary` volume named `scratch`. The files in this volume correspond to the same `scratch` folder used in previous versions of Compute.

Do not save your scripts or data files anywhere in your Compute container's file system except in "Storage" or "Temporary".

You can create up to 3 containers at the same time.

Create a container, select the HEASARC compute image (e.g. HEASARCV6.35.1), select a user volume, and finally make sure HEASARC data is checked under data volumes.

SciServer

Compute

Interactive Notebooks

Jobs

Created At

2025-04-04 19:57:38.0

Create container

Important Information about Compute Node

File System

Storage

persistent

Temporary

scratch

Most of the

Use Storage

By default, a

Use Temporary

By default, a

Do not save your scripts

Create a new container

Container name

SWIFT

Domain

Interactive Docker Compute Domain

Shared Intel Xeon E7 systems. All containers are limited to 100GiB of RAM. Unused containers are shut down after 3 days.

Compute Image

HEASARCV6.35.1

Contains Heasoft packages and software. Based on Ubuntu 22.04

User volumes

☐ All

☐ XMM-workshop, Storage Volume created by rjtanner

☒ persistent, Storage Volume created by navaneethpk

☒ scratch, Temporary Volume created by navaneethpk

Data volumes

☐ All

☐ AstroPath Data Public

☐ GALAHDR4

☐ Getting Started

☒ HEASARC data

☐ HathiTrust

☐ Manga

☐ Ocean Circulation

☐ Poseidon

☐ Recount

☐ SDSS Associated Data

☐ SDSS DAS

☐ SDSS DR9 Imaging

☐ SDSS SAS

☐ SDSS Spectra

☐ Turbulence (ceph)

☐ Turbulence (filedb)

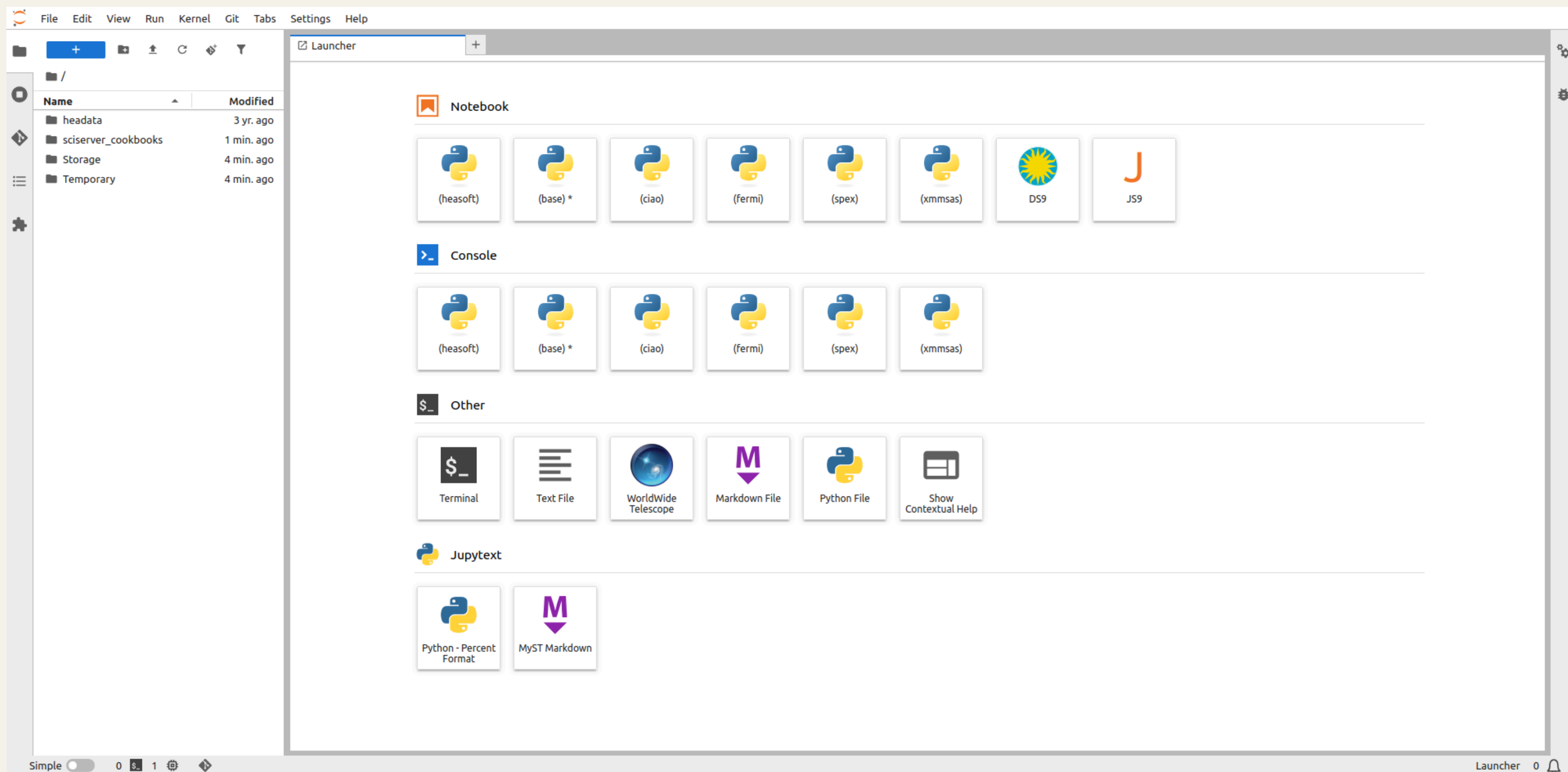
Create

Powered by:

NST

idies

JOHNS HOPKINS UNIVERSITY



File systems

There are two types of storage areas: Temporary and Persistent.

- The content of the **persistent** area persists between sessions, and can be accessed from multiple containers. It has a limit of 10GB.

`/home/idies/workspace/Storage/<username>/persistent/`

- The temporary area has larger disk space, but its content is not guaranteed to persist between sessions.

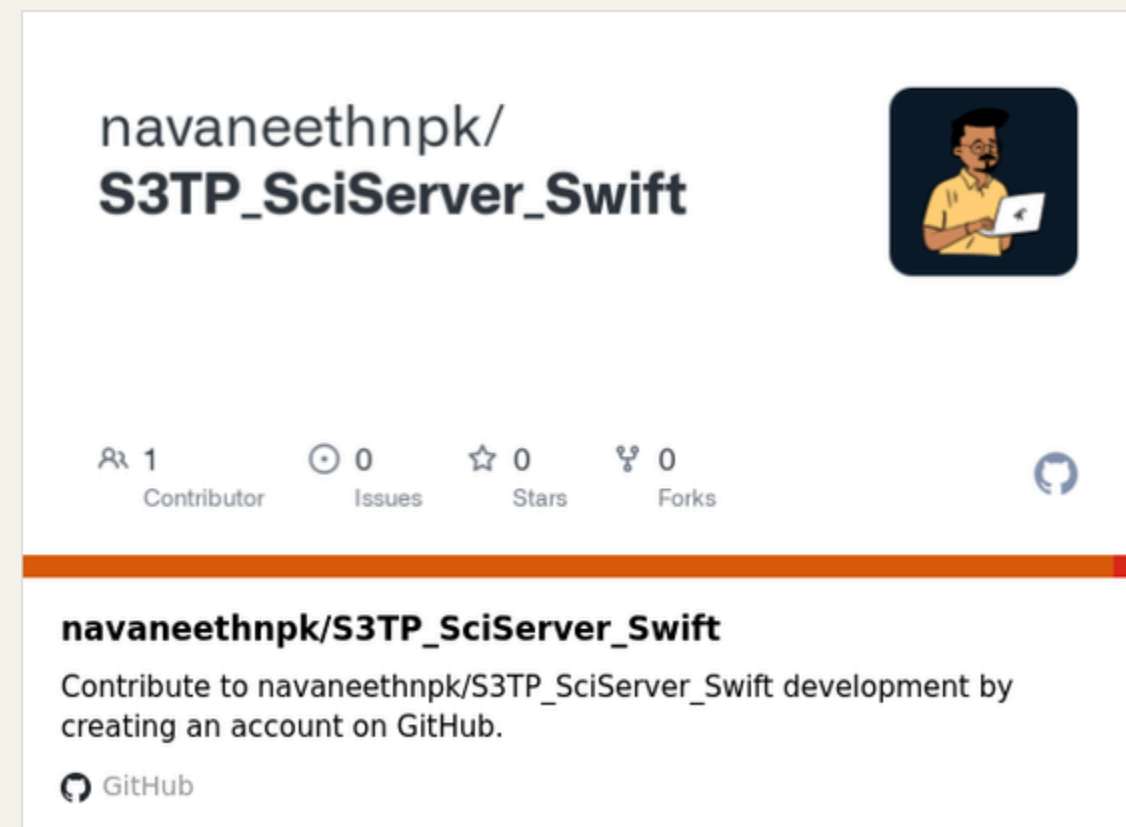
`/home/idies/workspace/Temporary/<username>/scratch/`

headata: This is typically a read-only folder that contains datasets provided by HEASARC.

sciserver_cookbooks: This contains a set of tutorial notebooks and example workflows.

Storage: This is your safe, persistent, and backed-up storage area.

Temporary: This is your non-backed-up temporary storage, ideal for large or intermediate files.



https://github.com/navaneethnpk/S4TP_SciServer_Swift

ESA Datalabs [016.0/BETA]

ESA Datalabs is available as "Public Moderated Beta"
If you wish to apply for access, please [submit your request here](#).
We have identified an issue in COSMOS that may prevent access to EDL after registration approval. Our team is working to resolve it.
If you're affected, please submit a [support ticket](#) so we can assist you.



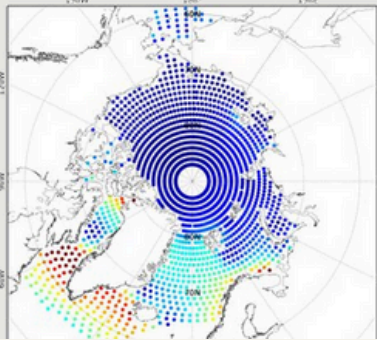
**«YOU CAN EITHER MOVE YOUR QUESTIONS
OR THE DATA. [...] OFTEN IT TURNS OUT
TO BE MORE EFFICIENT TO MOVE THE
QUESTIONS THAN TO MOVE THE DATA.»**

Jim Gray, eScience: A Transformed Scientific Method

BRING YOUR QUESTIONS TO THE DATA

There is a new paradigm, opening completely new opportunities for discovery – a data-intensive approach to science. In many domains, we have entered what could be called the golden age of surveys, with several large-scale projects, spanning decades, between finished, ongoing, and planned activities. ESA is responsible, or is a major partner, in several of these initiatives.

There is, however, a new profound change: data has become a major technological challenge. Increases by multiple orders of magnitude in dataset size means that transferring data to a scientist is often unfeasible.



<https://datalabs.esa.int>

ESA Datalabs is an online platform created by the European Space Agency (ESA). It helps users work with space data more easily. Instead of downloading large files, users can run their code directly on ESA's servers.